

Postnikov Towers

A Postnikov tower for a path-connected space X is a commutative diagram as at the right, such that:

- (1) The map $X \rightarrow X_n$ induces an isomorphism on π_i for $i \leq n$.
- (2) $\pi_i(X_n) = 0$ for $i > n$.

As we saw in Example 4.17, every connected CW complex X has a Postnikov tower, and this is unique up to homotopy equivalence by Corollary 4.19.

If we convert the map $X_n \rightarrow X_{n-1}$ into a fibration, its fiber F_n is a $K(\pi_n X, n)$, as is apparent from a brief inspection of the long exact sequence of homotopy groups for the fibration:

$$\pi_{i+1}(X_n) \rightarrow \pi_{i+1}(X_{n-1}) \rightarrow \pi_i(F_n) \rightarrow \pi_i(X_n) \rightarrow \pi_i(X_{n-1})$$

We can replace each map $X_n \rightarrow X_{n-1}$ by a fibration $X'_n \rightarrow X'_{n-1}$ in succession, starting with $X_2 \rightarrow X_1$ and working upward. For the inductive step we convert the composition $X_n \rightarrow X_{n-1} \hookrightarrow X'_{n-1}$ into a fibration $X'_n \rightarrow X'_{n-1}$ fitting into the commutative diagram at the right. Thus we obtain a Postnikov tower satisfying also the condition

$$\begin{array}{c} \vdots \\ \downarrow \\ X_3 \\ \downarrow \\ X_2 \\ \downarrow \\ X_1 \\ \uparrow \\ X \end{array}$$

$$\begin{array}{ccc} X_n & \hookrightarrow & X'_n \\ \downarrow & & \downarrow \\ X_{n-1} & \hookrightarrow & X'_{n-1} \end{array}$$

- (3) The map $X_n \rightarrow X_{n-1}$ is a fibration with fiber a $K(\pi_n X, n)$.

To the extent that fibrations can be regarded as twisted products, up to homotopy equivalence, the spaces X_n in a Postnikov tower for X can be thought of as twisted products of Eilenberg-MacLane spaces $K(\pi_n(X), n)$.

For many purposes, a CW complex X can be replaced by one of the stages X_n in a Postnikov tower for X , for example if one is interested in homotopy or homology groups in only a finite range of dimensions. However, to determine the full homotopy type of X from its Postnikov tower, some sort of limit process is needed. Let us investigate this question in somewhat greater generality.

Given a sequence of maps $\cdots \rightarrow X_2 \rightarrow X_1$, define their **inverse limit** $\varprojlim X_n$ to be the subspace of the product $\prod_n X_n$ consisting of sequences of points $x_n \in X_n$ with x_n mapping to x_{n-1} under the map $X_n \rightarrow X_{n-1}$. The corresponding algebraic notion is the inverse limit $\varprojlim X_n$ of a sequence of group homomorphisms $\cdots \rightarrow G_2 \rightarrow G_1$, which is the subgroup of $\prod_n G_n$ consisting of sequences of elements $g_n \in G_n$ with g_n mapping to g_{n-1} under the homomorphism $G_n \rightarrow G_{n-1}$.

Proposition 4.67. *For an arbitrary sequence of fibrations $\cdots \rightarrow X_2 \rightarrow X_1$ the natural map $\lambda: \pi_i(\varprojlim X_n) \rightarrow \varprojlim \pi_i(X_n)$ is surjective, and λ is injective if the maps $\pi_{i+1}(X_n) \rightarrow \pi_{i+1}(X_{n-1})$ are surjective for n sufficiently large.*

Proof: Represent an element of $\varprojlim \pi_i(X_n)$ by maps $f_n: (S^i, s_0) \rightarrow (X_n, x_n)$. Since the projection $p_n: X_n \rightarrow X_{n-1}$ takes $[f_n]$ to $[f_{n-1}]$, by applying the homotopy lifting

property for the pair (S^i, s_0) we can homotope f_n , fixing s_0 , so that $p_n f_n = f_{n-1}$. Doing this inductively for $n = 2, 3, \dots$, we get $p_n f_n = f_{n-1}$ for all n simultaneously, which gives surjectivity of λ .

For injectivity, note first that inverse limits are unaffected by throwing away a finite number of terms at the end of the sequence of spaces or groups, so we may assume the maps $\pi_{i+1}(X_n) \rightarrow \pi_{i+1}(X_{n-1})$ are surjective for all n . Given a map $f: S^i \rightarrow \varprojlim X_n$, suppose we have nullhomotopies $F_n: D^{i+1} \rightarrow X_n$ of the coordinate functions $f_n: S^i \rightarrow X_n$ of f . We have $p_n F_n = F_{n-1}$ on S^i , so $p_n F_n$ and F_{n-1} are the restrictions to the two hemispheres of S^{i+1} of a map $g_{n-1}: S^{i+1} \rightarrow X_{n-1}$. If the map $\pi_{i+1}(X_n) \rightarrow \pi_{i+1}(X_{n-1})$ is surjective, we can rechoose F_n so that the new g_{n-1} is nullhomotopic, that is, so that $p_n F_n \simeq F_{n-1} \text{ rel } S^i$. Applying the homotopy lifting property for (D^{i+1}, S^i) , we can make $p_n F_n = F_{n-1}$. Doing this inductively for $n = 2, 3, \dots$, we see that $f: S^i \rightarrow \varprojlim X_n$ is nullhomotopic and λ is injective. $\square$

One might wish to have a description of the kernel of λ in the case of an arbitrary sequence of fibrations $\dots \rightarrow X_2 \rightarrow X_1$, though for our present purposes this question is not relevant. In fact, $\text{Ker } \lambda$ is naturally isomorphic to $\varprojlim^1 \pi_{i+1}(X_n)$, where $\varprojlim^1$ is the functor defined in §3.F. Namely, if $f: S^i \rightarrow \varprojlim X_n$ determines an element of $\text{Ker } \lambda$, then the sequence of maps $g_n: S^{i+1} \rightarrow X_n$ constructed above gives an element of $\prod_n \pi_{i+1}(X_n)$, well-defined up to the choice of the nullhomotopies F_n . Any new choice of F_n is obtained by adding a map $G_n: S^{i+1} \rightarrow X_n$ to F_n . The effect of this is to change g_n to $g_n + G_n$ and g_{n-1} to $g_{n-1} - p_n G_n$. Since $\varprojlim^1 \pi_{i+1}(X_n)$ is the quotient of $\prod_n \pi_{i+1}(X_n)$ under exactly these identifications, we get $\text{Ker } \lambda \approx \varprojlim^1 \pi_{i+1}(X_n)$. Thus for each $i > 0$ there is a natural exact sequence

$$0 \rightarrow \varprojlim^1 \pi_{i+1}(X_n) \rightarrow \pi_i(\varprojlim X_n) \rightarrow \varprojlim \pi_i(X_n) \rightarrow 0$$

The proposition says that the $\varprojlim^1$ term vanishes if the maps $\pi_{i+1}(X_n) \rightarrow \pi_{i+1}(X_{n-1})$ are surjective for sufficiently large n .

Corollary 4.68. *For the Postnikov tower of a connected CW complex X the natural map $X \rightarrow \varprojlim X_n$ is a weak homotopy equivalence, so X is a CW approximation to $\varprojlim X_n$.*

Proof: The composition $\pi_i(X) \rightarrow \pi_i(\varprojlim X_n) \xrightarrow{\lambda} \varprojlim \pi_i(X_n)$ is an isomorphism since $\pi_i(X) \rightarrow \pi_i(X_n)$ is an isomorphism for large n . $\square$

Having seen how to decompose a space X into the terms in its Postnikov tower, we consider now the inverse process of building a Postnikov tower, starting with X_1 as a $K(\pi, 1)$ and inductively constructing X_n from X_{n-1} . It would be very nice if the fibration $K(\pi, n) \rightarrow X_n \rightarrow X_{n-1}$ could be extended another term to the right, to form a fibration sequence

$$K(\pi, n) \rightarrow X_n \rightarrow X_{n-1} \rightarrow K(\pi, n+1)$$

for this would say that X_n is the homotopy fiber of a map $X_{n-1} \rightarrow K(\pi, n+1)$, and homotopy classes of such maps are in one-to-one correspondence with elements of $H^{n+1}(X_{n-1}; \pi)$ by Theorem 4.57. Since the homotopy fiber of $X_{n-1} \rightarrow K(\pi, n+1)$ is the same as the pullback of the path fibration $PK(\pi, n+1) \rightarrow K(\pi, n+1)$, its homotopy type depends only on the homotopy class of the map $X_{n-1} \rightarrow K(\pi, n+1)$, by Proposition 4.62. Note that the last term $K(\pi, n+1)$ in the fibration sequence above cannot be anything else but a $K(\pi, n+1)$ since its loop space must be homotopy equivalent to the first term in the sequence, a $K(\pi, n)$.

In general, a fibration $F \rightarrow E \rightarrow B$ is called **principal** if there is a commutative diagram

$$\begin{array}{ccccc} F & \longrightarrow & E & \longrightarrow & B \\ \downarrow & & \downarrow & & \downarrow \\ \Omega B' & \longrightarrow & F' & \longrightarrow & E' \longrightarrow B' \end{array}$$

where the second row is a fibration sequence and the vertical maps are weak homotopy equivalences. Thus if all the fibrations in a Postnikov tower for X happen to be principal, we have a diagram as at the right, where each X_{n+1} is, up to weak homotopy equivalence, the homotopy fiber of the map $k_n: X_n \rightarrow K(\pi_{n+1}, n+2)$. The map k_n is equivalent to a class in $H^{n+2}(X_n; \pi_{n+1}(K))$ called the n^{th} **k -invariant** of X . These classes specify how to construct X inductively from Eilenberg–MacLane spaces. For example, if all the k_n 's are zero, X is just the product of the spaces $K(\pi_n X, n)$, and in the general case X is some sort of twisted product of $K(\pi_n X, n)$'s.

$$\begin{array}{ccccc} & & \vdots & & \\ & & \downarrow & & \\ K(\pi_3 X, 3) & \longrightarrow & X_3 & \xrightarrow{k_3} & K(\pi_4 X, 5) \\ & & \downarrow & & \\ K(\pi_2 X, 2) & \longrightarrow & X_2 & \xrightarrow{k_2} & K(\pi_3 X, 4) \\ & & \downarrow & & \\ K(\pi_1 X, 1) = X_1 & \xrightarrow{k_1} & & & K(\pi_2 X, 3) \end{array}$$

To actually build a space from its k -invariants is usually too unwieldy a procedure to be carried out in practice, but as a theoretical tool this procedure can be quite useful. The next result tells us when this tool is available:

Theorem 4.69. *A connected CW complex X has a Postnikov tower of principal fibrations iff $\pi_1(X)$ acts trivially on $\pi_n(X)$ for all $n > 1$.*

Notice that in the definition of a principal fibration, the map $F \rightarrow \Omega B'$ automatically exists and is a homotopy weak equivalence once one has the right-hand square of the commutative diagram with its vertical maps weak homotopy equivalences. Thus the question of whether a fibration is principal can be rephrased in the following way: Given a map $A \rightarrow X$, which one can always replace by an equivalent fibration if one likes, does there exist a fibration $F \rightarrow E \rightarrow B$ and a commutative square as at the right, with the vertical maps weak homotopy equivalences? By replacing A and X with CW approximations and converting the resulting map $A \rightarrow X$ into an inclusion via a mapping cylinder, the question becomes whether a CW pair (X, A) is equivalent to a fibration pair (E, F) , that

$$\begin{array}{ccc} A & \longrightarrow & X \\ \downarrow & & \downarrow \\ F & \longrightarrow & E \end{array}$$

is, whether there is a fibration $F \rightarrow E \rightarrow B$ and a map $(X, A) \rightarrow (E, F)$ for which both $X \rightarrow E$ and $A \rightarrow F$ are weak homotopy equivalences. In general the answer will rarely be yes, since the homotopy fiber of $A \hookrightarrow X$ would have to have the weak homotopy type of a loop space, which is a rather severe restriction. However, in the situation of Postnikov towers, the homotopy fiber is a $K(\pi, n)$ with π abelian since $n \geq 2$, so it is a loop space. But there is another requirement: The action of $\pi_1(A)$ on $\pi_n(X, A)$ must be trivial for all $n \geq 1$. This is equivalent to the action of $\pi_1(F)$ on $\pi_n(E, F)$ being trivial, which is always the case in a fibration since under the isomorphism $p_* : \pi_n(E, F) \rightarrow \pi_n(B, x_0)$ an element $\gamma\alpha - \alpha$, with $\gamma \in \pi_1(F)$ and $\alpha \in \pi_n(E, F)$, maps to $p_*(\gamma)p_*(\alpha) - p_*(\alpha)$ which is zero since $p_*(\gamma)$ lies in the trivial group $\pi_1(x_0)$.

The relative group $\pi_n(X, A)$ is always isomorphic to π_{n-1} of the homotopy fiber of the inclusion $A \hookrightarrow X$, so in the case at hand when the homotopy fiber is a $K(\pi, n)$, the only nontrivial relative homotopy group is $\pi_{n+1}(X, A) \approx \pi$. In this case the necessary condition of trivial action is also sufficient:

Lemma 4.70. *Let (X, A) be a CW pair with both X and A connected, such that the homotopy fiber of the inclusion $A \hookrightarrow X$ is a $K(\pi, n)$, $n \geq 1$. Then there exists a fibration $F \rightarrow E \rightarrow B$ and a map $(X, A) \rightarrow (E, F)$ inducing weak homotopy equivalences $X \rightarrow E$ and $A \rightarrow F$ iff the action of $\pi_1(A)$ on $\pi_{n+1}(X, A)$ is trivial.*

Proof: It remains only to prove the ‘if’ implication. As we noted just before the statement of the lemma, the groups $\pi_i(X, A)$ are zero except for $\pi_{n+1}(X, A) \approx \pi$. If the action of $\pi_1(A)$ on $\pi_{n+1}(X, A)$ is trivial, the relative Hurewicz theorem gives an isomorphism $\pi_{n+1}(X, A) \approx H_{n+1}(X, A)$. Since (X, A) is n -connected, we may assume A contains the n -skeleton of X , so X/A is n -connected and the absolute Hurewicz theorem gives $\pi_{n+1}(X/A) \approx H_{n+1}(X/A)$. Hence the quotient map $X \rightarrow X/A$ induces an isomorphism $\pi_{n+1}(X, A) \approx \pi_{n+1}(X/A)$ since the analogous statement for homology is certainly true.

Since $\pi_{n+1}(X/A) \approx \pi$, we can build a $K(\pi, n+1)$ from X/A by attaching cells of dimension $n+3$ and greater. This leads to the commutative diagram at the right, where the vertical maps are inclusions and the lower row is obtained by converting the map k into a fibration. The map $A \rightarrow F_k$ is a weak homotopy equivalence by the five-lemma applied to the map between the long exact sequences of homotopy groups for the pairs (X, A) and (E_k, F_k) , since the only nontrivial relative groups are π_{n+1} , both of which map isomorphically to $\pi_{n+1}(K(\pi, n+1))$. $\square$

$$\begin{array}{ccccc} A & \longrightarrow & X & \longrightarrow & X/A \\ \downarrow & & \downarrow \simeq & \searrow k & \downarrow \\ F_k & \longrightarrow & E_k & \longrightarrow & K(\pi, n+1) \end{array}$$

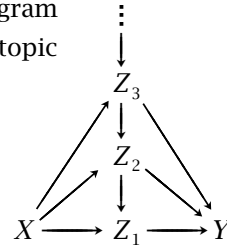
Proof of 4.69: In view of the lemma, all that needs to be done is identify the action of $\pi_1(X)$ on $\pi_n(X)$ with the action of $\pi_1(X_n)$ on $\pi_{n+1}(X_{n-1}, X_n)$ for $n \geq 2$, thinking of the map $X_n \rightarrow X_{n-1}$ as an inclusion. From the exact sequence

$$0 = \pi_{n+1}(X_{n-1}) \rightarrow \pi_{n+1}(X_{n-1}, X_n) \xrightarrow{\partial} \pi_n(X_n) \rightarrow \pi_n(X_{n-1}) = 0$$

we have an isomorphism $\pi_{n+1}(X_{n-1}, X_n) \approx \pi_n(X_n)$ respecting the action of $\pi_1(X_n)$. And the map $X \rightarrow X_n$ induces isomorphisms on π_1 and π_n , so we are done. $\square$

Let us consider now a natural generalization of Postnikov towers, in which one starts with a map $f: X \rightarrow Y$ between path-connected spaces rather than just a single space X . A **Moore–Postnikov tower** for f is a commutative diagram as shown at the right, with each composition $X \rightarrow Z_n \rightarrow Y$ homotopic to f , and such that:

- (1) The map $X \rightarrow Z_n$ induces an isomorphism on π_i for $i < n$ and a surjection for $i = n$.
- (2) The map $Z_n \rightarrow Y$ induces an isomorphism on π_i for $i > n$ and an injection for $i = n$.
- (3) The map $Z_{n+1} \rightarrow Z_n$ is a fibration with fiber a $K(\pi_n F, n)$ where F is the homotopy fiber of f .



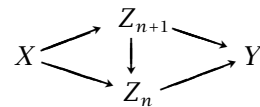
A Moore–Postnikov tower specializes to a Postnikov tower by taking Y to be a point and then setting $X_n = Z_{n+1}$, discarding the space Z_1 which has trivial homotopy groups.

Theorem 4.71. *Every map $f: X \rightarrow Y$ between connected CW complexes has a Moore–Postnikov tower, which is unique up to homotopy equivalence. A Moore–Postnikov tower of principal fibrations exists iff $\pi_1(X)$ acts trivially on $\pi_n(M_f, X)$ for all $n > 1$, where M_f is the mapping cylinder of f .*

Proof: The existence and uniqueness of a diagram satisfying (1) and (2) and commutative at least up to homotopy follows from Propositions 4.13 and 4.18 applied to the pair (M_f, X) with M_f the mapping cylinder of f . Having such a diagram, we proceed as in the earlier case of Postnikov towers, replacing each map $Z_n \rightarrow Z_{n-1}$ by a homotopy equivalent fibration, starting with $Z_2 \rightarrow Z_1$ and working upward. We can then apply the homotopy lifting property to make all the triangles in the left half of the tower strictly commutative. After these steps the triangles in the right half of the diagram commute up to homotopy, and to make them strictly commute we can just replace each map to Y by the composition through Z_1 .

To see that the fibers of the maps $Z_{n+1} \rightarrow Z_n$ are Eilenberg–MacLane spaces as in condition (3), consider two successive levels of the tower.

We may arrange that the maps $X \rightarrow Z_{n+1} \rightarrow Z_n \rightarrow Y$ are inclusions by taking mapping cylinders, first of $X \rightarrow Z_{n+1}$, then



of the new $Z_{n+1} \rightarrow Z_n$, and then of the new $Z_n \rightarrow Y$. From the left-hand triangle we see that $Z_{n+1} \rightarrow Z_n$ induces an isomorphism on π_i for $i < n$ and a surjection for $i = n$, hence $\pi_i(Z_n, Z_{n+1}) = 0$ for $i < n + 1$. Similarly, the other triangle gives $\pi_i(Z_n, Z_{n+1}) = 0$ for $i > n + 1$. To show that $\pi_{n+1}(Z_n, Z_{n+1}) \approx \pi_{n+1}(Y, X)$ we use the following diagram:

$$\begin{array}{ccccccccc}
\pi_{n+1}(Z_{n+1}) & \longrightarrow & \pi_{n+1}(Z_n) & \longrightarrow & \pi_{n+1}(Z_n, Z_{n+1}) & \longrightarrow & \pi_n(Z_{n+1}) & \longrightarrow & \pi_n(Z_n) \\
\downarrow = & & \downarrow \approx & & \downarrow & & \downarrow = & & \downarrow \\
\pi_{n+1}(Z_{n+1}) & \longrightarrow & \pi_{n+1}(Y) & \longrightarrow & \pi_{n+1}(Y, Z_{n+1}) & \longrightarrow & \pi_n(Z_{n+1}) & \longrightarrow & \pi_n(Y) \\
\uparrow & & \uparrow = & & \uparrow & & \uparrow \approx & & \uparrow = \\
\pi_{n+1}(X) & \longrightarrow & \pi_{n+1}(Y) & \longrightarrow & \pi_{n+1}(Y, X) & \longrightarrow & \pi_n(X) & \longrightarrow & \pi_n(Y)
\end{array}$$

The upper-right vertical map is injective and the lower-left vertical map is surjective, so the five-lemma implies that the two middle vertical maps are isomorphisms. Since the homotopy fiber of an inclusion $A \hookrightarrow B$ has π_i equal to $\pi_{i+1}(B, A)$, we see that condition (3) is satisfied.

The statement about a tower of principal fibrations can be obtained as an application of Lemma 4.70. As we saw in the previous paragraph, there are isomorphisms $\pi_{n+1}(Y, X) \approx \pi_{n+1}(Z_n, Z_{n+1})$, and these respect the action of $\pi_1(X) \approx \pi_1(Z_{n+1})$, so Lemma 4.70 gives the result. $\square$

Besides the case that Y is a point, which yields Postnikov towers, another interesting special case of Moore-Postnikov towers is when X is a point. In this case the space Z_n is an n -connected covering of Y , as in Example 4.20. The n -connected covering of Y can also be obtained as the homotopy fiber of the n^{th} stage $Y \rightarrow Y_n$ of a Postnikov tower for Y . The tower of n -connected coverings of Y can be realized by principal fibrations by taking Z_n to be the homotopy fiber of the map $Z_{n-1} \rightarrow K(\pi_n Y, n)$ that is the first nontrivial stage in a Postnikov tower for Z_{n-1} .

$$\begin{array}{c}
\vdots \\
\downarrow \\
Z_2 \rightarrow K(\pi_3 Y, 3) \\
\downarrow \\
Z_1 \rightarrow K(\pi_2 Y, 2) \\
\downarrow \\
Y \rightarrow K(\pi_1 Y, 1)
\end{array}$$

A generalization of the preceding theory allowing nontrivial actions of π_1 can be found in [Robinson 1972].

Obstruction Theory

It is very common in algebraic topology to encounter situations where one would like to extend or lift a given map. Obvious examples are the homotopy extension and homotopy lifting properties. In their simplest forms, extension and lifting questions can often be phrased in one of the following two ways:

The Extension Problem. Given a CW pair (W, A) and a map $A \rightarrow X$, does this extend to a map $W \rightarrow X$?

$$\begin{array}{ccc}
A & \longrightarrow & X \\
\downarrow & \nearrow \text{---} & \\
W & &
\end{array}$$

The Lifting Problem. Given a fibration $X \rightarrow Y$ and a map $W \rightarrow Y$, is there a lift $W \rightarrow X$?

$$\begin{array}{ccc}
& & X \\
& \nearrow \text{---} & \downarrow \\
W & \longrightarrow & Y
\end{array}$$

In order for the lifting problem to include things like the homotopy lifting property, it should be generalized to a relative form: